

Time Series Exercise Sheet 9

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Exercise 9.1

For the process $\mathbf{Y}_t = (Y_t^{(1)}, Y_t^{(2)})^T$ generated by

$$Y_t^{(j)} = X_t^{(j)} + \varepsilon_t, \quad j = 1, 2$$

with $X^{(1)}, X^{(2)}$ and ε independent and where ε is white noise. Write

$$\sigma_j^2 = \text{Var}(X_t^{(j)})$$

$$\sigma_\varepsilon^2 = \text{Var}(\varepsilon_t)$$

$$\tilde{\gamma}_\tau^{(1,2)} = \text{Cov}(X_{t+\tau}^{(1)}, X_t^{(2)})$$

Use $\gamma_\tau^{(1,2)}$ to denote the cross-covariance sequence between $\{Y_t^{(1)}\}$ and $\{Y_t^{(2)}\}$. Calculate the correlation between $\{Y_t^{(1)}\}$ and $\{Y_t^{(2)}\}$ at all lags.

Solution 9.1

Since ε_t is white noise, and noting that $X_{j,t}$ and ε_t are independent, we have

$$\begin{aligned} \gamma_\tau^{(1,2)} &= \text{Cov}(Y_{t+\tau}^{(1)}, Y_{t+\tau}^{(2)}) \\ &= \text{Cov}(X_{t+\tau}^{(1)}, X_t^{(2)}) + \text{Cov}(X_{t+\tau}^{(1)}, \varepsilon_t) + \text{Cov}(\varepsilon_{t+\tau}, X_t^{(2)}) + \text{Cov}(\varepsilon_{t+\tau}, \varepsilon_t) \\ &= \text{Cov}(X_{t+\tau}^{(1)}, X_t^{(2)}) + \text{Cov}(\varepsilon_{t+\tau}, \varepsilon_t) \\ &= \begin{cases} \tilde{\gamma}_0^{(1,2)} + \sigma_\varepsilon^2 & \text{if } \tau = 0, \\ \tilde{\gamma}_\tau^{(1,2)} & \text{if } \tau \neq 0, \end{cases} \end{aligned}$$

and marginally

$$\begin{aligned} \gamma_0^{(1,1)} &= \text{Cov}(Y_t^{(1)}, Y_t^{(1)}) \\ &= \text{Cov}(X_t^{(1)}, X_t^{(1)}) + \text{Cov}(\varepsilon_t, \varepsilon_t) \\ &= \sigma_1^2 + \sigma_\varepsilon^2, \end{aligned}$$

finally we have

$$\gamma_0^{(2,2)} = \sigma_2^2 + \sigma_\varepsilon^2.$$

Therefore,

$$\rho_\tau^{(1,2)} = \begin{cases} \frac{\tilde{\gamma}_0^{(1,2)} + \sigma_\varepsilon^2}{\sqrt{\sigma_1^2 + \sigma_\varepsilon^2} \sqrt{\sigma_2^2 + \sigma_\varepsilon^2}} & \text{if } \tau = 0, \\ \frac{\tilde{\gamma}_\tau^{(1,2)}}{\sqrt{\sigma_1^2 + \sigma_\varepsilon^2} \sqrt{\sigma_2^2 + \sigma_\varepsilon^2}} & \text{otherwise.} \end{cases}$$

Exercise 9.2

For the delay process, with $\alpha > 0$ and $\nu \in \mathbb{Z}$

$$X_t^{(1)} = \alpha X_{t-\nu}^{(2)} + \varepsilon_t, \quad t \in \mathbb{Z}$$

where $\{\varepsilon_t\}$ is white noise and $\{X_t^{(2)}\}$ is stationary and uncorrelated with $\{\varepsilon_t\}$ at all lags

- (a) Calculate the correlation between $\{X_t^{(1)}\}$ and $\{X_t^{(2)}\}$ at all lags.
- (b) Calculate the cross spectrum, and determine the amplitude (absolute value) and phase (argument) of the cross-spectrum.

Solution 9.2

- (a) Again noting that $X_t^{(j)}$ and ε_t are independent, we have

$$\begin{aligned} \gamma_\tau^{(1,2)} &= \text{Cov} \left(X_{t+\tau}^{(1)}, X_t^{(2)} \right) \\ &= \alpha \text{Cov} \left(X_{t+\tau-\nu}^{(2)}, X_t^{(2)} \right) + \text{Cov} \left(\varepsilon_{t+\tau}, X_t^{(2)} \right) \\ &= \alpha \text{Cov} \left(X_{t+\tau-\nu}^{(2)}, X_t^{(2)} \right) \\ &= \alpha \gamma_{\tau-\nu}^{(2,2)}, \end{aligned}$$

additionally,

$$\begin{aligned} \gamma_0^{(1,1)} &= \alpha^2 \sigma_2^2 + \sigma_\varepsilon^2, \\ \gamma_0^{(2,2)} &= \sigma_2^2. \end{aligned}$$

Therefore,

$$\begin{aligned} \rho_\tau^{(1,2)} &= \frac{\alpha \gamma_{\tau-\nu}^{(2,2)}}{\sqrt{\alpha^2 \sigma_2^2 + \sigma_\varepsilon^2} \sqrt{\sigma_2^2}} \\ &= \frac{\rho_{\tau-\nu}^{(2,2)}}{\sqrt{1 + \sigma_\varepsilon^2 / (\alpha \sigma_2)^2}} \end{aligned}$$

- (b) We have

$$\begin{aligned} S_{1,2}(f) &= \sum_{\tau=-\infty}^{\infty} \gamma_\tau^{(1,2)} e^{-2\pi i f \tau} \\ &= \alpha \sum_{\tau=-\infty}^{\infty} \gamma_{\tau-\nu}^{(2,2)} e^{-2\pi i f \tau} \\ &= \alpha e^{-2\pi i f \nu} \sum_{\tau=-\infty}^{\infty} \gamma_\tau^{(2,2)} e^{-2\pi i f \tau} \\ &= \alpha S_{2,2}(f) e^{-2\pi i f \nu} \end{aligned}$$

showing that the amplitude is the spectral density of $X_t^{(2)}$ scaled by α and the phase is $f \mapsto -2\pi f \nu$.

Exercise 9.3

For $\mathbf{Y}_t = \begin{pmatrix} Y_t^{(1)} \\ Y_t^{(2)} \end{pmatrix} \sim N(\mathbf{0}, \mathbf{\Sigma})$ independently in t with $\mathbf{\Sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}$, determine the expectation and variance of $\hat{\gamma}_\tau^{(1,2)}$ in

$$\hat{\mathbf{\Gamma}}_\tau = \begin{bmatrix} \hat{\gamma}_\tau^{(1,1)} & \hat{\gamma}_\tau^{(1,2)} \\ \hat{\gamma}_\tau^{(2,1)} & \hat{\gamma}_\tau^{(2,2)} \end{bmatrix} = \frac{1}{N} \sum_{t=1}^{N-|\tau|} \mathbf{Y}_{t+\tau} \mathbf{Y}_t^T$$

Solution 9.3

$$\begin{aligned}
\mathbb{E} \left[\hat{\gamma}_\tau^{(1,2)} \right] &= \mathbb{E} \left[\frac{1}{N} \sum_{t=1}^{N-|\tau|} Y_{t+\tau}^{(1)} Y_t^{(2)} \right] \\
&= \frac{1}{N} \sum_{t=1}^{N-|\tau|} \mathbb{E} \left[Y_{t+\tau}^{(1)} Y_t^{(2)} \right] \\
&= \begin{cases} \sigma_{12} & \tau = 0 \\ 0 & \tau \neq 0. \end{cases}
\end{aligned}$$

$$\begin{aligned}
\text{Var}(\hat{\gamma}_\tau^{(1,2)}) &= \text{Var} \left(\frac{1}{N} \sum_{t=1}^{N-|\tau|} Y_{t+\tau}^{(1)} Y_t^{(2)} \right) \\
&= \mathbb{E} \left[\left(\frac{1}{N} \sum_{t=1}^{N-|\tau|} Y_{t+\tau}^{(1)} Y_t^{(2)} \right)^2 \right] - \mathbb{E} \left[\hat{\gamma}_\tau^{(1,2)} \right]^2 \\
&= \frac{1}{N^2} \left\{ \sum_{t=1}^{N-|\tau|} \mathbb{E} \left[\left(Y_{t+\tau}^{(1)} \right)^2 \left(Y_t^{(2)} \right)^2 \right] + \sum_{t \neq s}^{N-|\tau|} \mathbb{E} \left[Y_{t+\tau}^{(1)} Y_t^{(2)} Y_{s+\tau}^{(1)} Y_s^{(2)} \right] \right\} - \{ \mathbb{E}(\hat{\gamma}_\tau^{(1,2)}) \}^2.
\end{aligned}$$

Note that $\mathbb{E} \left[\left(Y_t^{(1)} \right)^2 \left(Y_t^{(2)} \right)^2 \right] = \sigma_{11}\sigma_{22} + 2\sigma_{12}^2$ since both $Y_t^{(1)}$ and $Y_t^{(2)}$ are Gaussian. Therefore, if $\tau = 0$,

$$\begin{aligned}
\text{Var}(\hat{\gamma}_0^{(1,2)}) &= \frac{1}{N^2} \left\{ \sum_{t=1}^N \mathbb{E} \left[\left(Y_t^{(1)} \right)^2 \left(Y_t^{(2)} \right)^2 \right] + \sum_{t \neq s}^N \mathbb{E} \left[Y_t^{(1)} Y_t^{(2)} Y_s^{(1)} Y_s^{(2)} \right] \right\} - \sigma_{12}^2 \\
&= \frac{1}{N^2} \left\{ \sum_{t=1}^N \mathbb{E} \left[\left(Y_t^{(1)} \right)^2 \left(Y_t^{(2)} \right)^2 \right] + \sum_{t \neq s}^N \mathbb{E} \left[Y_t^{(1)} Y_t^{(2)} Y_s^{(1)} Y_s^{(2)} \right] \right\} - \sigma_{12}^2 \\
&= \frac{1}{N} (\sigma_{11}\sigma_{22} + 2\sigma_{12}^2) + \frac{N-1}{N} \sigma_{12}^2 - \sigma_{12}^2 \\
&= \frac{1}{N} (\sigma_{11}\sigma_{22} + \sigma_{12}^2).
\end{aligned}$$

If $\tau \neq 0$,

$$\begin{aligned}
\text{Var}(\hat{\gamma}_\tau^{(1,2)}) &= \frac{1}{N^2} \left\{ \sum_{t=1}^{N-|\tau|} \mathbb{E} \left[\left(Y_{t+\tau}^{(1)} \right)^2 \left(Y_t^{(2)} \right)^2 \right] + \sum_{t \neq s}^{N-|\tau|} \mathbb{E} \left[Y_{t+\tau}^{(1)} Y_t^{(2)} Y_{s+\tau}^{(1)} Y_s^{(2)} \right] \right\} \\
&= \frac{1}{N^2} \{ (N - |\tau|) \sigma_{11}\sigma_{22} + 0 \} \\
&= \frac{N - |\tau|}{N^2} \sigma_{11}\sigma_{12}.
\end{aligned}$$

In summary,

$$\begin{aligned}
\mathbb{E} \left[\hat{\gamma}_\tau^{(1,2)} \right] &= \sigma_{12} \delta_\tau, \\
\text{Var}(\hat{\gamma}_\tau^{(1,2)}) &= \frac{N - |\tau|}{N^2} (\sigma_{11}\sigma_{22} + \sigma_{12}^2 \delta_\tau),
\end{aligned}$$

where

$$\delta_\tau = \begin{cases} 1 & \tau = 0, \\ 0 & \tau \neq 0. \end{cases}$$